Literature Review: Modern Real-Time Payment Architectures & Lakehouse Paradigms

1. Context & Foundational Knowledge

Modern financial engineering relies heavily on real-time event streaming coupled with reliable analytical data stores. Traditional monolithic database systems struggle to handle high-concurrency payment transactions while simultaneously serving heavy analytical queries (OLAP). Modern data engineering standards emphasize the Medallion Lakehouse Architecture (Bronze, Silver, Gold layers) to ensure event idempotency, data deduplication, and decoupled streaming/batch processing.

Data Layer	Primary Responsibility	Data Integrity Level	Target Consumers
Bronze Layer	Raw ingestion of append-only stream and batch payloads	Unvalidated historical record	System audit, raw replays
Silver Layer	Schema validation, event deduplication, state matching	Cleaned, enforced, unified	Analytics engineers, automated pipelines
Gold Layer	Aggregated star-schemas and dimensional financial metrics	High assurance (Ledger-ready)	BI tools, executive reporting, ML models

2. Analysis of Existing Solutions & Limitations (Research Gap)

Existing payment processing and reconciliation frameworks suffer from structural vulnerabilities that limit scalability and threaten financial precision:

- **Coupled Operational and Analytical Workloads:** Querying real-time payment states directly from OLTP databases degrades system performance and checkout speeds, leading to increased latency during peak transactional volumes.
- **Handling Late and Duplicate Events:** Webhook retries and network partitions frequently result in duplicate deliveries or out-of-order events that break naive aggregation queries, corrupting financial metrics.

- **Manual 3-Way Reconciliation:** Reconciling processor logs against internal ledgers and bank statements is traditionally manual, slow, and prone to human error, introducing significant audit risk.

3. Proposed Engineering Solution & Justification (PayLens)

PayLens bridges these structural gaps through an advanced hybrid architecture:

1. **Hybrid Ingestion & Medallion Storage:** Combines streaming events (Event Hubs) and batch data (ADF) into an immutable Bronze layer, transitioning seamlessly to validated Delta Lake (Silver) and structured dimensional models (Gold).
2. **Idempotent Processing & Deduplication:** Implements strict `event_id` tracking and Delta MERGE logic to safely handle duplicates, late arrivals, and out-of-order states without corrupting historical ledgers.
3. **Automated Settlement & Reconciliation:** Embeds append-only financial ledger rules and a 3-way automated matching module to eliminate reconciliation gaps and secure audit trails.